



## Reflect with us - Week 6



This assignment will not be graded and is only for practice.

Suppose the matrix representation of a linear transformation  $T: R^3 \rightarrow R^3$  with respect to ordered bases

$$\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\} \quad \text{for the domain and}$$

$$\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\} \quad \text{for the range,}$$

is  $I_{3 \times 3}$ , i.e., the identity matrix of order 3.

Let  $AA$  denote the matrix representation of the linear transformation  $T$  with respect to the standard ordered basis of  $R^3$  for both domain and range. Which of the following are true?

a.  $A = I_{3 \times 3}$ , i.e., identity matrix of order 3.

b.  $AA$  is a singular matrix.

c.  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$

d.  $\det(A) = 1$

e.  $\det(A) = -1$

f.  $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

### Solution:

Observe that, in question, you are asked to find the matrix representation of  $T$  with respect to the standard ordered bases of  $R^3$  for both domain and range. So you have to find the image of each element of the standard ordered basis of  $R^2$  via the linear transformation  $T$ .

Note:  $I_{3 \times 3}$  is matrix representation of the  $T$  with respect to the ordered bases  $\beta$  and  $\gamma$ .

Step 0:

What is  $T(1, 0, 1)$ ? (Enter your answer in the form of a row vector in  $R^3$ )

**Note:** The answer will be of the form (a, b, c) where a, b and c are in  $R$

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

From the given data, we can calculate

$$T(1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 0(1, 0, 1) = (1, 0, 0)$$

**Accepted Answers:**



(Type: String) (1, 0, 0)

(Type: String) (1,0,0)

1 point

What is  $T(0, 1, 0)T(0, 1, 0)$ ?**No, the answer is incorrect.****Score: 0****Feedback:**

From the given data, we can calculate

$$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)$$

**Accepted Answers:**

(Type: String) (0, 1, 0)

(Type: String) (0,1,0)

1 point

What is  $T(0, 0, 1)T(0, 0, 1)$ ?**No, the answer is incorrect.****Score: 0****Feedback:**

From the given data, we can calculate

$$T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)$$

**Accepted Answers:**

(Type: String) (1, 0, 1)

(Type: String) (1,0,1)

1 point

Step 1: Step 1:

Now we have to find the image of  $(1, 0, 0)(1, 0, 0)$  via the linear transformation  $T$ . We know the image of each element of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$   $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ . So at first we have to write  $(1, 0, 0)(1, 0, 0)$  in terms of the elements of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$   $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ .

$$(1, 0, 0) = \alpha_1(1, 0, 1) + \beta_1(0, 1, 0) + \gamma_1(0, 0, 1)(1, 0, 0) = \alpha_1(1, 0, 1) + \beta_1(0, 1, 0) + \gamma_1(0, 0, 1)$$

(0, 0, 1)

What is the value of  $\alpha_1$ ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 1

1 point

What is the value of  $\beta_1$ ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of  $\gamma_1$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) -1

1 point

1 point

Now, can you find  $T(1, 0, 0)T(1, 0, 0)$ ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

From the given data, we can calculate

$$T(1, 0, 0) = T(1(1, 0, 1) + 0(0, 1, 0) - 1(0, 0, 1))T(1, 0, 0) = 1T(1, 0, 1) + 0T(0, 1, 0) - 1T(0, 0, 1)$$

$$T(1, 0, 0) = T(1(1, 0, 1) + 0(0, 1, 0) - 1(0, 0, 1))T(1, 0, 0) = 1T(1, 0, 1) + 0T(0, 1, 0) - 1T(0, 0, 1)$$

Here we know the images of  $(1, 0, 1)$ ,  $(0, 1, 0)$ ,  $(1, 0, 1)$ ,  $(0, 1, 0)$  and  $(0, 0, 1)$ .

$$T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) - 1(1, 0, 1) = (0, 0, -1)T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) - 1(1, 0, 1) = (0, 0, -1)$$

**Accepted Answers:**

Yes

No

Step 2:Step 2:

Now we have to find the image of  $(0, 1, 0)$  via the linear transformation  $T$ . We know the image of each element of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ . So at first we have to write  $(0, 1, 0)$  in terms of the elements of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ .

$$(0, 1, 0) = \alpha_2(1, 0, 1) + \beta_2(0, 1, 0) + \gamma_2(0, 0, 1)(0, 1, 0) = \alpha_2(1, 0, 1) + \beta_2(0, 1, 0) + \gamma_2(0, 0, 1)$$

What is the value of  $\alpha_2$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of  $\beta_2$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

What is the value of  $\gamma_2 \gamma_2$ ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

Now, can you find  $T(0, 1, 0)T(0, 1, 0)$ ?☐ Yes☐ No**No, the answer is incorrect.****Score: 0****Feedback:**

$$T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))$$

$$T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)$$

$$T(0, 1, 0) = (0, 1, 0)T(0, 1, 0) = (0, 1, 0)$$

**Accepted Answers:**

Yes

No

Step 3:Step 3:

Now we have to find the image of  $(0, 0, 1)(0, 0, 1)$  via the linear transformation  $T$ . We know the image of each element of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ . So at first we have to write  $(0, 0, 1)(0, 0, 1)$  in terms of the elements of the basis  $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ .

$$(0, 0, 1) = \alpha_3(1, 0, 1) + \beta_3(0, 1, 0) + \gamma_3(0, 0, 1)(0, 0, 1) = \alpha_3(1, 0, 1) + \beta_3(0, 1, 0) + \gamma_3(0, 0, 1)$$

What is the value of  $\alpha_3$ ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of  $\beta_3$ ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of  $\gamma_3$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Now, can you find  $T(0, 0, 1)T(0, 0, 1)$ ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

$$T(0, 0, 1) = T(0(1, 0, 1) + 0(0, 1, 0) + 1(0, 0, 1))T(0, 0, 1) = T(0(1, 0, 1) + 0(0, 1, 0) + 1(0, 0, 1))$$

$$T(0, 0, 1) = 0T(1, 0, 1) + 0T(0, 1, 0) + 1T(0, 0, 1)T(0, 0, 1) = 0T(1, 0, 1) + 0T(0, 1, 0) + 1T(0, 0, 1)$$

$$T(0, 0, 1) = (1, 0, 1)T(0, 0, 1) = (1, 0, 1)$$

**Accepted Answers:**

Yes

No

Step 4: Step 4: Now we have to find out the matrix  $AA$ . For that we have to write down the images with respect to the standard ordered basis.

$$T(1, 0, 0) = 0(1, 0, 0) + 0(0, 1, 0) - 1(0, 0, 1)T(1, 0, 0) = 0(1, 0, 0) + 0(0, 1, 0) - 1(0, 0, 1)$$

$$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$$

$$T(0, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)T(0, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$$

Write the coefficients column wise to get the matrix  $AA$ .

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Step 5: Step 5:

In this step, you have to find the determinant of the matrix  $AA$ .

1 point

Can you find the  $\det(A)$ ?

☐ Yes

☐ No

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Yes

No

- Expanding with respect to the first row:

$$\det(A) = 0(1 - 0) - 0(0 - 0) + 1(0 + 1)\det(A) = 0(1 - 0) - 0(0 - 0) + 1(0 + 1)$$

- Expanding with respect to the first column:

$$\det(A) = 0(1 - 0) - 0(0 - 0) - 1(0 - 1)\det(A) = 0(1 - 0) - 0(0 - 0) - 1(0 - 1)$$

- You can expand with respect to any other rows or column.

Enter the value of *determinant of the matrix A* determinant of the matrix A

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

*1 point*

Your score is: 0/17